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AI Interview Trainer Agent

Exploring the Power of Agentic AI with IBM Granite and IBM Bob

Artificial Intelligence

Generative AI

Agentic AI • RAG

Domain: Artificial Intelligence & Generative AI

Project Title: Interview Trainer Agent

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THE CHALLENGE

Fragmented, Generic Interview Preparation

Job seekers often struggle to prepare for interviews because questions, role expectations, technical topics, and HR guidance are scattered across different sources. Generic interview preparation does not consider a candidate's profile, experience level, resume, and target job role — making it difficult to practice effectively and build real confidence.

THE OBJECTIVE

Build an AI-powered Interview Trainer Agent using **IBM Granite** and **RAG** that delivers truly personalized interview preparation.

- 1

Profile-Based Preparation — tailored to experience, resume & target role.
- 2

RAG-Based Retrieval — pulls technical, behavioral & HR knowledge from a curated base.
- 3

Personalized Question Generation — role-specific technical, HR & behavioral questions.
- 4

Model Answers & Feedback — sample answers, improvement tips & response feedback.
- 5

Multi-Role Support — adapts to different job roles & experience levels.
- 6

Interview Assessment — evaluates relevance, clarity, technical depth & communication.
- 7

IBM Granite Integration — via IBM Cloud for AI-powered generation & analysis.
- 8

Interactive Practice — real-time practice sessions with personalized guidance.

OVERVIEW

AI-Powered Interview Trainer Agent

Built using **IBM Bob**, **IBM watsonx.ai**, and **IBM Granite**, the solution delivers personalized, interactive interview preparation across job roles. A Retrieval-Augmented Generation (RAG) pipeline retrieves relevant questions, role expectations, behavioral scenarios, and guidance from a curated knowledge base — enabling accurate, context-aware responses for both technical and non-technical roles, based on the user's profile, experience, skills, and resume.

HOW IT WORKS



FOUR KEY AGENT CAPABILITIES

- Interview Question Agent**

Generates role-specific technical, HR, behavioral, and situational questions.
- Answer Evaluation Agent**

Scores responses and surfaces strengths, weaknesses, and improvement suggestions.
- Interview Guidance Agent**

Provides model answers, preparation strategies, and role-specific tips.
- RAG Knowledge Agent**

Retrieves relevant interview knowledge to keep responses accurate and grounded.

TECHNOLOGY STACK



Outcome: An end-to-end intelligent interview preparation assistant that helps users practice realistic interviews, identify weaknesses, improve their answers, and build confidence for competitive job interviews — powered by Agentic AI rather than a simple Q&A chatbot.